

Unshackled

GSD390 Final Project in Unreal Engine 5 with C++

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Version: 1.0

Table of Contents

- 1. [Title and Overview](#)
 - 2. [Gameplay Mechanics](#)
 - 3. [Technical Requirements](#)
 - 4. [Level Design and World Structure](#)
 - 5. [Visual and Audio Design](#)
 - 6. [Story and Narrative](#)
 - 7. [User Interface \(UI\) and HUD](#)
 - 8. [Testing and Iteration Plan](#)
 - 9. [Project Timeline](#)
-

1. Title and Overview

1.1 Game Title

The working title of this game is *Unshackled*.

1.2 Game Genre

Mystery/Fantasy RPG. Turn-based + A-RPG

1.3 Target Audience

Attribute	Details
Age Group / Rating	ESRB T, 13 and up
Platform	Windows, Linux
Player Demographic	Catered to players who enjoy turn-based combat and beat-em ups, as well as following a story
Play Session Length	3-4 hours

1.4 Game Summary

Taking place in a dungeon, the game aims to curate an experience that combines both turn-based combat mechanics and A-RPG style beat-em ups as a means to progress through the game, while also following a visual-novel based story. Players will progress through the game fighting waves of enemy per level, unlocking story-content along the way, and engaging in turn-based combat with mini-boss and boss-level entities.

The story follows Sirius. Plagued with amnesia, they can't figure out why they are locked in a dungeon, better yet, why they are in the depths of said dungeon housing the most dangerous of the dangerous, all while missing their left arm. In finding out the injustice that takes place in the dungeon, they set off to break out of the dungeon, rallying allies to join their party along the way. Can Sirius break out of the unforgiving hellscape that surrounds them and their allies, and can they remember why they are there in the first place?

1.5 Unique Selling Points

- Floors are real-time A-RPG fights against waves of enemies, and bosses are turn-based. How the player plays decides the Retribution bonuses and starting conditions for the boss fight.
- Grounding lets the player drop allies around the floor as semi-independent units that cast their own spells and interact with each other. This turns wave combat into positioning and area control and rewards player for strategizing instead of button-mashing.
- The last fight is against Sirius, using every skill the player unlocked. Upgrading your character over the whole game is also building your final opponent, so the twist hits both in the story and in the gameplay.
- Story pieces unlock floor by floor, so the player's progress in the escape is also their progress in finding out who Sirius is and why they were locked up at the bottom of the dungeon.

2. Gameplay Mechanics

2.1 Core Mechanics

Core Loop:

- Defeat waves of enemies
- Gain EXP + items
- Upgrade party members
- Unlock story content
- Engage in turn-based combat with boss
- Repeat

Mechanic	Description	Purpose in Gameplay
Basic movement	Simple navigation around levels	Movement will be a core part of the overworld aspect of the game. Mobs of enemies will chase the player and party around the level in which the player's party needs to be able to maneuver strategically around to defeat enemies efficiently

Mechanic	Description	Purpose in Gameplay
Grounding allies	Deploy members of party around the level to automatically interact with enemies. Party members will have unique interactions with enemies and allies when "grounded" (ex. casting specific spells or skills).	This serves as a way for the player to strategize throughout the level
Retribution	Sequence of party member order will affect turn-based combat by giving player's party additional bonuses (damage boosts, additional shields/heals, enemy debuffs, etc)	To reward players for strategy and planning when fighting enemies
Leader	Party members can switch a "leader" of party to start combat. This has unique effects and interactions with enemies and allies	This has unique effects and interactions with enemies and allies, both negative and positive

Objectives:

- Successfully escaping the dungeon.
- Recruiting new party members to aid in their dungeon escape.
- Regaining the lost memories of the protagonist.

Challenges:

- Enemy mobs (dungeon guards, wardens, dungeon floor bosses)
- Dungeon terrain

2.2 Controls and Player Input

Input Method(s): Keyboard & mouse, controller (Xbox / Playstation)

Input System: Unreal's Enhanced Input System, with two Input Mapping Contexts: **IMC_Overworld** for real-time floors and **IMC_TurnBased** for boss fights. The Player Controller can swap between them when combat changes mode.

'O' indicates overworld mechanics and 'T' indicates turn-based combat mechanics. Otherwise, action is a system-wide mechanic.

Action	Keyboard/Mouse	Controller
O-Move	WASD	Left Stick
O-Look	Move mouse	Right Stick
O-Jump	Space	Cross / A
Pause	Esc	Menu/Options

Action	Keyboard/Mouse	Controller
O-Physical Skill	X	X / Square
O-Special Skill	C	Y / Triangle
O-Ultimate Skill	V	RT / R2
Use item	Q	D-Pad up
O-Tackle	Left shift	B / Circle
O-Ground party member	G	LT / L2
Interact	Left mouse click	LB / L1
Open inventory	R	View / Touchpad
O-Switch leader	Tab + Scroll wheel (to cycle)	RB / R1 + D-pad (to select)
T-Toggle skills	Arrow keys	D-pad
T-Use skill	Space	Cross / A
T-Hover party member/Hover enemy	WASD	Left Stick

2.3 Player Progression

Stage / Level	New Ability or Mechanic	Challenge Introduced
Floor 1	Movement, Grounding, Retribution, Leader + all other core game mechanics + first party member	Waves of enemy mobs + floor boss + story mystery
Floor 2	Second party member + story progression	mobs + boss + terrain
Floor 3	Third party member + story progression	mobs + boss + terrain
Floor 4	fourth and final party member + story progression	mobs + boss + terrain
Floor 5	story progression	mobs + 2 boss fights
Escape	finale of story	betrayal (final boss fight)

2.4 AI Behavior

Enemy / NPC	Movement	Attacks / Actions	Decision-Making
Basic guard	Running (slightly faster than player's party)	basic melee attacks	Chases at start of level, does not yield until death
Wardens	slow walk	Slower but stronger melee attacks, can buff enemies	Spawns in a bit later, chases throughout level, does not yield until death

Enemy / NPC	Movement	Attacks / Actions	Decision-Making
Caster guards	levitation (slightly slower than player's party)	magic spells (can apply debuffs)	Follows player party until they are in range, runs away if player party is too close, stops after death
Caster wardens	levitation	very far ranged attacks (applies debuffs), can buff enemies	Casts spells towards player's party after spawning in, does not stop until defeated
Hounds	Sprints (much faster than player's party)	small ranged, powerful melee attacks	slowly follows player party trail(s) until spotted in vision, where it then hunts player down until death
Floor director (miniboss)	dependent on floor (each director is unique)	dependent on floor, miniboss will either be physical (short range melee), magic (long range spells), or both	dependent on miniboss, but they do not yield until defeated
Floor boss (Goat, serpent, lion, dragon, Chimera)	no movement	dependent on floor, each boss is unique	bosses will have turns equal to number of party members, and will choose most optimal moves given circumstances; stops once defeated
S. Ravana + Chimera	no movement	Final boss is our main protagonist, Sirius; will have all skills player unlocked that can be used to attack player party	Given three turns (the total number of remaining party members) and will choose most optimal moves given circumstances; stops once defeated

3. Technical Requirements

3.1 Unreal Engine 5.8 Features

Unshackled is top-down 2D pixel art with illustrated dialogue portraits. Nanite and Lumen are left off, since they don't help a sprite-based game.

Feature / Tool	Use
C++	Core systems: characters, combat, stats, turn-based battles, spawning, NPC, dialogue, saving
Blueprints	Character/enemy setup, level scripting, UI layout, tuning
Paper2D	Sprites, Flipbooks (animation), and Tile Maps (dungeon floors)
PaperZD	Animation state machines with directional facing; frame-based events for hit timing

Feature / Tool	Use
Orthographic Camera	Top-down follow camera at pixel scale
Enhanced Input	All controls, with overworld and turn-based mapping contexts
AI (Behavior Trees, Perception, NavMesh)	Enemy and grounded-ally behavior, sight detection, pathfinding
Gameplay Tags	Skill types, damage types, and status effects
Data Assets / Data Tables	Stats, skills, enemies, waves, EXP curves, and dialogue scripts
UMG + Common UI	HUD, menus, inventory, battle UI, dialogue box; controller menu navigation
UMG Animations	Portrait slide-in/out, speaker highlighting, and small reactions
Niagara	Sprite-based spell, hit, and ambient effects
Level Sequencer	Short story scenes and boss intros
MetaSounds / Sound Classes	Music transitions and volume sliders
SaveGame	Progress, party, inventory, story flags
Lighting	Lit sprites with point lights; Lumen and Nanite disabled
Chaos Physics	Collision/hitboxes and Tackle knockback only
Texture Settings	Pixel art uses nearest-neighbor filtering, no mipmaps; portraits use UI filtering. Anti-aliasing, motion blur, and auto-exposure off

3.2 Programming Requirements (C++)

C++ Class	Parent Class	Responsibility
AUnshackledCharacterBase	APaperZDCharacter	Shared base for party and enemies
APartyMemberCharacter	AUnshackledCharacterBase	Leader effects, grounding, following
AEnemyCharacter	AUnshackledCharacterBase	Drops, AI setup, enemy type data
AUnshackledPlayerController	APlayerController	Input contexts, leader switching, grounding, menus
AUnshackledGameMode	AGameModeBase	Waves, win/lose, switching to boss fights
UUnshackledGameInstance	UGameInstance	Party, inventory, and story flags across floors

C++ Class	Parent Class	Responsibility
AWaveSpawner	AActor	Spawns enemy waves from Data Tables
ATurnBattleManager	AActor	Turn order, Retribution bonuses, skill resolution
UBossDecisionComponent	UActorComponent	Scores and picks the best boss move
AEnergyAIController	AAIController	Behavior Trees and Perception
UBTTask_* / UBTService_*	UBTTaskNode / UBTService	Custom AI actions (keep distance, track, buff)
UHealthComponent	UActorComponent	HP, shields, damage, death
UStatsComponent	UActorComponent	Level, EXP, stats
USkillComponent	UActorComponent	Skills and cooldowns in both combat modes
UStatusEffectComponent	UActorComponent	Bufs and debuffs
UInventoryComponent	UActorComponent	Items
USkillData / UCharacterData	UPrimaryDataAsset	Skill and character data, including portrait sets
UDialogueManager	UGameInstanceSubsystem	Runs dialogue from Data Tables, sets story flags
UDialogueWidget	UUserWidget	Text reveal, portrait expressions, speaker highlight
UUnshackledSaveGame	USaveGame	Save data
UUnshackledHUDWidget / UBattleWidget	UUserWidget	HUD and battle UI data

Key Systems to Implement:

- ☐ 2D top-down character controller
- ☐ Real-time combat (hitboxes, damage, knockback, cooldowns)
- ☐ Grounding and party following
- ☐ Leader switching
- ☐ Wave spawning and floor progression
- ☐ Turn-based battles with Retribution
- ☐ Enemy AI and boss AI
- ☐ Bufs/debuffs
- ☐ EXP and upgrades
- ☐ Inventory
- ☐ Visual novel dialogue

- ☐ HUD, battle UI, menus
- ☐ Save/load

C++ vs. Blueprint Split: Game rules and systems are in C++, so they're fast, easier to debug (in my opinion), and should be able to merge cleanly between machines. Blueprints can handle the presentation side of things with character setup, animations, UI layout, level scripting, and tuning. Stats, skills, and dialogue will live in Data Tables so changes don't need a recompile.

3.3 Assets and Tools

Asset Type	Source	Custom / Premade	Notes
Sprites & Animations (Pixel)	Aseprite; itch.io / Fab packs	Mix	Main characters custom; generic enemies edited from packs
Tilesets (Pixel)	itch.io / Fab	Mostly premade	Custom tiles for hazards and floor features
Dialogue Portraits (Illustrated)	Krita	Custom	Bust portraits with multiple expressions
UI (Pixel)	Aseprite	Custom	HUD, icons, menus, dialogue frame
VFX Textures	FX packs + custom	Mix	Used in Niagara
Sound Effects	Freesound.org / Fab / itch.io	Premade	Slashes, spells, hits, footsteps, UI
Music	Waveform Free / Ardour + guitar	Custom	Original soundtrack featuring recorded guitar
Fonts	Free pixel fonts	Premade	Chosen for readability

Development Environment (Windows 11 + Fedora Linux):

Tool	Windows 11	Fedora
Unreal Engine 5.8	Epic Games Launcher	Epic's Linux build
IDE	Rider	Rider
Compiler	MSVC	Clang (UE toolchain)
Graphics API	DirectX 12	Vulkan
Pixel Art / Illustration	Aseprite / Krita	Aseprite / Krita
Audio	Waveform Free, Audacity	Ardour, Audacity
Version Control	Git + GitHub + LFS	Git + GitHub + LFS

3.4 Target Platform

These are only estimates and need to be confirmed by profiling builds on both OSes.

Requirement	Minimum	Recommended
OS	Windows 10 / 64-bit Linux	Windows 11 / current 64-bit Linux
CPU	i5-6400 / Ryzen 3 1200	i5-8400 / Ryzen 5 3600
GPU	GTX 960 / RX 470 (2 GB)	GTX 1060 / RX 580 (4 GB+)
RAM	8 GB	16 GB
Storage	5 GB	5 GB (SSD)
Graphics API	DX12 / Vulkan	DX12 / Vulkan
Input	Keyboard & mouse	Controller

Target Performance: 60 FPS at 1080p (recommended), 60 FPS at 720p (minimum).

4. Level Design and World Structure

4.1 World Layout

The overall world/setting of the game will be a multi-floor dungeon. The dungeon is deep underground, and is shaped like an upside-down trapezoidal prism (floors get bigger and longer as the player and party progresses upward).

Level / Area	Description	Key Features
Main dungeon space	Any space that is not a cell or safe spot on a given floor	This is where enemies will spawn into (mobs, miniboss, floor boss)
Cells	Cell to house prisoners	Can include loot, NPC interaction (where recruiting party members can take place, as well as basic dialogue with other NPCs)
Safe spots	Small, guard-free rooms tucked into each floor (old storage rooms, collapsed cells)	Save points, healing, party management (upgrades, leader/order changes), and optional dialogue with allies

Example Floor Layout:

4.2 Environmental Features

- Terrain: Can be setup in a way to challenge player movement; there will also be pillars that can be broken in a way to have floors collapse
- Buildings / Structures:
- Interactive Elements: doors, switches for certain gates or doors, pickups like items or keys
- Hazards: falling platforms (collapsed floors), enemy altered terrain

4.3 Flow and Pacing

As the game progresses, the game will become harder in a roughly linear manner. Floors progressively challenge player mechanics and strategy with enemies growing in power, and skills.

Example Difficulty Curve:



5. Visual and Audio Design

5.1 Art Style

- **Graphical Style:** 2d, 16-bit sprites
- **Color Palette:** Grey, brown, black, orange, yellow, earthy
- **Lighting & Mood:** Darker and broody
- **Influences / References:** *Persona* series, *The Mageseeker: A League of Legends Story*, *The Legend of Zelda: Majora's Mask*, *Slay the Spire 1 & 2*, *Genshin Impact*, *Puzzle & Dragons*

5.2 Character Design

Character	Appearance	Key Animations	Behavior / Personality
Sirius	Disheveled, long ivory hair, bandages covering torso, beat-up capri-like pants, barefoot. Red eyes.	Idle, Running, Jump, Attack, Death, leader, retribution, grounded	Because player plays as Sirius, dialogue options will reflect player's interpretation of Sirius's personality, with choices ranging from pessimistic, optimistic, and neutral. Sirius at the end of the day is charismatic and good at rallying his allies together. Throughout the story, he is constantly

Character	Appearance	Key Animations	Behavior / Personality
			juxtaposed with comparisons to a demon, though him nor his allies fully understand why; this in turn slowly makes him doubt his innocence.
Mal'Kholm Ecks	Tall, big, muscular, bald, and shirtless, with a brown, braided beard. Tan skin and yellow eyes. Big chains latches to both wrists, with a shunk of stone connected on each end of the chains. Short 5-inch inseam shorts, big brolic legs and barefoot	Idle, Running, Jump, Attack, Death, leader, retribution, grounded	He is a gentle, thoughtful, and patient man. Although portrayed as a bloodthirsty savage, he remains resolute in breaking out of the dungeon and reuniting with the revolution
Kalia	Short, black hair pixie cut. She is tall and lean. Green eyes. Dark skin complexion, she wears a cape over her baggy prison attire.	Idle, Running, Jump, Attack, Death, leader, retribution, grounded	She is passionate, prideful, and bold in all regard. She deeply cares for her allies, and is the type to bleed for them.
Hanuboy	A small pika. They are shackled with a big leg iron at least three-times are big as he is.	Idle, Running, Jump, Attack, Death, leader, retribution, grounded	Very sassy. Often frustrated that his only means of communicating is through squeaks, despite fully understanding human language.
Basic guard	Basic, plated attire, similar to knights from the medieval times. All wield a basic broadsword.	Running, attack, death	No distinct personality other than to take down player's party.
Wardens	heavily armored in a rustic gold. Built similar to conventional 'juggernaut' archetypes. All wield a shield and heavy mace.	Walking, attack, death	No distinct personality other than to take down player's party.
Caster guards	Draped with a brown robe that covers everything but their glowing yellow eyes. All wield a magic staff.	Running, attack, death	Cowardly in nature, becomes confident when out of range of player's party.
Caster wardens	Draped with a navy robe, they encompass themselves in magic circles. Their face and	Wind up, attack, death	They are maniacal and will anything to utterly defeat the player's party

Character	Appearance	Key Animations	Behavior / Personality
	body are also hidden, other than their gross grin.		
Floor director (floor miniboss)	three types: physical damage based ones have a lot of armor, more than wardens, and ride rams into combat. Magic based ones are enamored with magic circles, a and robe that fully covers their sillhoutte. Hybrid ones (magic + physical) ride battle rams and are drenched in a magic circles and a cloak.	riding, wind up, attack, death	Almost robotic. No feelings and thoughts, just there to take down the player and party.
Hounds	Wolves who jaws are replaced with giant serrated ones. Hounds are black outside of their distinctly blue eyes and chrome/white jaws.	sprints, sniffing, attack, death	A ruthless beast, whos consumed only by bloodlust
Floor boss	In order, the bosses will have traditional goat, serpent, lion, dragon, and finally a chimera (the four come together and fuse) features, with them all having slight variations in design to reflect their abilities.	Idle, attack, death	All bosses are stoic, and calm. Their only goal is to serve their master (revealed to be S. Ravana) and will not stop at any cost.
S. Ravana (formerly Sirius)	Clean, upkept. Slick back hair, with white-glowing tribal markings all over his body. His left arm is one composed of magic. Dressed in a robe. Their eyes are blindfolded, though their red eyes shine right through. Veiled by powerful wind magic. Despite wearing sandals, they only float.	Idle, attack, death	Arrogant, cunning, charismatic, and ruthless. Their former memories with their former allies breaking out of the dungeon does nothing but digusts them.

5.3 Audio Design

- **Sound Effects:** Attacks (spells, physical attacks, buffs, debuffs), picking up items, using items, death, certain doors/gates, switches, retribution, grounding, clicks, more to come...
- **Music:** Combat music and passive music throughout game aims to be opposites of each other, with calming ambient music during non-combat scenes, and much more intense and thrilling sonic

landscape during combat.

- **Voice Acting:** No voice acting will be in this game.
- **How Audio Enhances Gameplay:** Audio changes based on stage of game: default battle music, changes to enhanced battle music when mini-boss spawns, changes to floor boss music, and "lore" music plays when story progresses. Final fight will have it's own unique theme. Music in this game will try to invoke the feelings of the party. Sound effects will also be distinct, and almost jarring to the music, to better highlight it's desired effect.

6. Story and Narrative

6.1 Plot Summary

- **Setting:** Inside a 5 floor dungeon.
- **Main Plot:**
- **Key Characters:** Sirius, Mal'Kholm Ecks, Kalia, Hanuboy

6.2 Objectives

- **Main Objective:** To break out of dungeon rooted in corruption and cruelty, and to change the system that is complacent in this mistreatment
- **Player Motivation:**
- **Secondary Objectives:** Learn the mystery behind Sirius's identity

6.3 Dialogue and Cutscenes

- **Dialogue System:** Visual novel text boxes with illustrated portraits and expressions. Sirius has dialogue choices (optimistic / neutral / pessimistic) that change ally reactions but not the main plot.
- **Cutscenes:** Short in-engine scenes made with Level Sequencer for major story beats; all other story moments use the dialogue system.

Cutscene / Story Beat	Trigger	Content
Awakening	Game start	Sirius wakes in a cell at the bottom of the dungeon, missing an arm, with no memories
Recruitment scenes	Entering an ally's cell (Floors 1–3)	Each ally is introduced and joins the party
Boss intros	Entering a boss chamber	Each boss appears and pledges loyalty to "the master"
Memory flashes	Defeating each floor boss	Short, cryptic flashbacks from Sirius's past
The name	Floor 4	The party learns the name S. Ravana. Clues from throughout the game should start to make sense for the party
The chimera	Floor 5	The four bosses fuse into the chimera

Cutscene / Story Beat	Trigger	Content
Betrayal	Reaching the exit	Sirius's memories return; he becomes S. Ravana and turns on the party, and revives Chimera to fight the party
Ending	Defeating S. Ravana + Chimera	The party escapes with Mal'Kholm Ecks and heads toward the reuniting with the revolution

7. User Interface (UI) and HUD

7.1 UI Elements

Menus:

- **Main Menu:** New game, load game, settings, quit
- **Pause Menu:** Save, quit, settings
- **Settings Menu:** Audio, graphics, controls
- **Game Over / Victory Screen:** Rewards (exp gained, items, skills unlocked) <- if floor cleared, otherwise, a menacing game-over screen that gives the player an option to return to last save, restart floor, or quit to menu.

HUD: 'O' indicates overworld elements and 'T' indicates turn-based combat elements.

HUD Element	Screen Position	Information Shown
O-Party Health Bar	Above player party (hovers above heads regardless of position)	current health / total combined health
O-HUD bar	Bottom middle of screen	Character picture, skills & cooldowns, item equipped
O-Current character leading picture	Left side inside HUD bar	Picture of current leader of party
O-Ability / Ability cooldowns	right side inside HUD bar	skills equiped and cooldowns if used (picture of skill will be greyed out with timer until back up)
O-Half-circle mini-map	Top center of screen	portion of map party is in, mobs indicated by red marker
O-Grounded allies list	bottom left corner	Current health and resource bar
O-Retribution bar	Bottom right corner	Sequence order in character switches need to happen
T-Character health bars	hover above character position on screen	Current health / Total health

HUD Element	Screen Position	Information Shown
T-Character resource bars	hover above character position on screen, right below health bar	Current resource / Total resource
T-Hud bar	bottom middle of screen	options to open skill menu, block, or open item menu
T-retribution progress	bottom right of screen	Progression for retribution and next order sequence

7.2 UX Design

UI aims to keep information crucial to player success compact and easy to glance at while in combat, and thorough and in-depth during turn-based combat. This allows for hack-n-slash sequences to be smooth and turn-based combat satisfying to players. HUD elements will also be customizable to players as they will be able to expand or reduce sizing to their liking. There will also be clear and distinct visual effects like damage numbers and small shakes (ability to toggle will be available) to indicate severity of damage, as well as status conditions having clear contrastable colors. Interactables will also be clearly highlighted on the map.

For example red = damage, green = heals, blue = shields, purple/pink = debuffs.

8. Testing and Iteration Plan

8.1 Testing Strategy

Test Type	Method	Frequency
Functional / Bug Testing	Test each feature in a small test map right after implementation	Every feature
Automated Tests	Unreal Automation Tests for C++ logic (damage math, status effects, Retribution bonuses, turn order and etc.)	Every build
Cross-Platform Builds	Package and run on both Windows 11 and Fedora	Weekly
Performance Testing	Unreal Insights, with max possible waves of enemies on screen	During every milestones
Playtesting	friends play a build and fill out a short feedback form	Three rounds that being the prototype, alpha, and beta stages
Balance Testing	Adjust stats and enemy waves in Data Tables based on the playtesting results	After each of the playtests

Bug Tracking: Cna use GitHub Issues, which is labeled by severity and platform the Windows / Linux

8.2 Iteration Plan

After each playtest, feedback is sorted into bugs, balance, and design changes. Crashes and blockers are fixed first, then balance through data tables/sheets, then any design changes. If time runs short, content can always be cut before the systems, so the build always stays complete and playable.

Scope Tiers:

- **Must Have (MVP):** Floors 1–2 fully playable; Sirius and 2 allies; grounding, leader, and Retribution; 3 enemy types; 2 turn-based boss fights; dialogue system; HUD and menus; save/load
- **Should Have:** Floors 3–5, all the possible enemy types, the chimera fight, all 3 allies, full soundtrack
- **Nice to Have:** Final S. Ravana fight, dialogue choices that affect ally reactions, packaged Linux release

9. Project Timeline

9.1 Development Phases

Phase	Dates	Tasks / Milestones	Deliverable
Setup & Prototyping	Sep 30 – Oct 14	Repo & cross-platform setup, movement, real-time combat, one enemy with AI, grounding, using placeholder art	Playable gray-box floor (should be done by Oct 18)
Core Systems	Oct 14 – Nov 8	Turn-based battle handler, Retribution implementation, leader, enemy waves, stat + EXP, dialogue, HUD	Alpha: Floor 1 complete (Nov 8)
Content	Nov 9 – Nov 25	Floor 2+, more enemies and bosses, sprites, portraits, music	Beta: MVP content (Nov 25)
Polish & Testing	Nov 26 – Dec 7	Playtest fixes, balance, VFX/SFX, optimization (will have a lighter load week over Thanksgiving just so I can enjoy some Turkey heheh)	Release candidate (Dec 7)
Final Build	Dec 10 – Dec 14	Packaged Windows + Linux builds, final bug fixes, submission	Final build (Dec 14)

Playtest Rounds: Oct 18 (prototype), Nov 8 (alpha), Nov 25 (beta)

Task Allocation

Solo project; all programming, art, audio, and design by me, Lyons Tran 🤖

Final note

I know that this is a very ambitious project, and doing so by myself is certainly going to be hard given I've never engaged in extensive game development. However, I am confident in my abilities to finish most of my intended vision by the final deadline, December 14th. I have an extensive coding background, so I believe feature implementation and debugging is an area I'm very comfortable in, and any areas I'm lacking in, I am good at looking through documentation. More than that, I've had a deep appreciation of video games since I

can remember, and given the freedom I have this semester (I have a very light load), I am planning on going all out into this project.

I am very excited and I can't wait for you to monitor my progression, and try out my game when the time comes.
